# Organizing Fiscal Capacity*

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**Abstract**

This paper investigates how the organization of a tax authority's presence across the national territory impacts fiscal capacity. I examine a reform of the Brazilian tax authority, which led to a more centralized structure with fewer, larger local offices. I show that I show that tax revenues decrease by 3.9% in municipalities previously served by a closed office. The effects are stronger in areas with less third-party reporting, where local information is more valuable. Distance acts as a friction, making it harder to gather local information. After the reform, the increased distance from tax offices results in lower tax revenues and fewer visits by tax agents.

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## 1 — Introduction

Governments in low- and middle-income countries struggle to increase their tax revenues (Besley and Persson 2013, 2014). One way to overcome this challenge is to enhance the effectiveness of tax authorities, the agencies in charge of tax administration. Recent work has emphasized how their effectiveness is impacted by the quality of tax agents and by the incentives they face (Bergeron et al. 2022; Khan, Khwaja, and Olken 2016). However, even though these agencies are often large and complex bureaucracies, little is known about how the organization of their presence across the national territory impacts fiscal capacity.

An important decision facing tax authorities is the choice between a centralized structure, with few local offices, and a more decentralized structure, with many local offices. A decentralized structure may enable tax agents to monitor taxpayers more effectively (Scott 1998; Lee and Zhang 2017) and better detect evasion. Beyond the obvious increase in administrative costs, a decentralized structure can reduce the tax authority's effectiveness when there are agency problems or when agents are not equally able to handle complex tax cases. In these circumstances, a centralized structure may allow the tax authority to more easily monitor agents to ensure they exert effort and do not misbehave. Similarly, a centralized structure may reduce communication costs and make it easier for tax agents to obtain help from more knowledgeable colleagues when handling complex cases.

In order to shed light on the costs of a centralized structure, this paper studies a reform of the Brazilian tax authority, the Federal Revenue of Brazil. I leverage variation in how different areas were affected by this reform to examine how a more centralized structure impacts fiscal capacity. First, I show that tax collection declines in areas previously served by a local office that was closed due to the reform. I then provide evidence suggesting that having fewer offices makes it harder to collect local economic information. I document how physical distance between tax agents and taxpayers acts as a barrier to acquiring this information

The Brazilian tax authority is an ideal context to study the costs associated with a centralized structure. This tax authority employs more than 20,000 officials and is tasked with tax administration over a territory of over 8,510,000 km$^2$. It is responsible for the collection of around 55% of the government tax revenues, including income tax, social security contributions, and taxes on firms'

inputs and profits. Following mandated government budget cuts, which led to a hiring freeze, a decline in personnel, and a reduction in the number of maintainable managerial positions, the authority underwent a significant reorganization in 2020. This reform closed one-fourth of the local offices responsible for tax enforcement. Municipalities served by the closed offices were reassigned to the jurisdiction of some of the remaining offices. These offices also absorbed most of the tax agents from the closed locations, making the reform a *de facto* centralization.

To empirically investigate the impact of this centralization, I combine various data sources to track the evolution of the Federal Revenue of Brazil presence and effectiveness across the national territory over the period 2016 to 2022. I reconstruct changes of tax offices' locations and tax jurisdictions boundaries from the internal regulations of the tax authority. I measure tax authority effectiveness using information on the amount of revenues it collects in each of the 5,565 municipalities - the lowest administrative level in Brazil. Moreover, I leverage novel administrative data tracing tax authority employees across offices and their trips for work. The resulting dataset allows me to measure the effects of the reform at a geographically disaggregated level.

In the first part of the paper, I provide evidence on the costs, in terms of tax enforcement, associated with a more centralized structure. My difference-in-differences design compares the evolution of tax revenues between municipalities whose jurisdiction tax office closed - *closing* municipalities - and municipalities whose jurisdiction tax office did not close - *not closing* municipalities. I find that *closing* municipalities experience a 3.9% reduction in federal tax collection. The effect is stark and persists up to the end of the sample period in 2022. There is no evidence of differential pre-trends in tax collection; this lends credibility to the identification assumption of parallel trends. Moreover, I employ various measures to document that there is no evidence of differential decline in economic activity - either before or after the closings. This confirms that the decline in tax collection is due to lower tax enforcement and not to a decline in the tax base. I also show that there is no decline in local taxes collected by the municipalities themselves - a useful placebo as one would not expect these taxes to be influenced by federal tax enforcement.

Having shown that the reform reduces tax enforcement in *closing* municipalities, I now investigate what drives this effect. While tax agents' ability to detect evasion using technology and hard data should be unaffected by their location, their ability to gather local information may

be. Therefore, we might expect stronger effects of office closings on enforcement in areas where local information is more valuable. Local information is particularly relevant in settings with less third-party reporting (Jensen 2022). In Brazil, inspections by the Ministry of Labor ensure that firms register their employees, creating an information trail on employees' income and firms' activities. This helps tax agents detect evasion even without local information. I use microdata on labor inspections from the years before the reform to classify municipalities based on the intensity of third-party reporting during that period. Heterogeneity analysis shows that office closings have a stronger effect in municipalities where third-party reporting is less prevalent, and local information is more valuable

Why do the closings make it harder to collect local information? One possibility is that tax officials are now farther from the municipalities they need to monitor, and this distance makes it more difficult to gather local information. To investigate the role of distance, I first document that the reform increases the travel time between a *closing* municipality and the jurisdiction's tax office by an average of 50 minutes. I then use an instrumental variable approach to isolate the effect of distance on tax enforcement.

My empirical strategy leverages the baseline variation in distance between a municipality and the closest *other* office within the same fiscal region. This distance does not affect tax enforcement before the reform, as what matters is the actual distance to the jurisdiction's tax office. However, for *closing* municipalities, this distance becomes the lower bound on the actual distance to the jurisdiction's tax office after the reform. Using this approach, I find that municipalities that end up one hour farther from a tax office experience a 1.7% reduction in tax revenues.

I then investigate how distance worsens tax officials' ability to collect *local information*. One possibility is that distance acts as a friction, preventing tax officials from physically visiting areas that are farther away. To explore this, I analyze the number of trips made by tax officials from a local office to a municipality. I find that distance reduces the number of trips tax officials make to a municipality. Moreover, these trips tend to last longer and cost more. Distance increases the cost of visiting municipalities both in terms of agents' time and monetary expenses.

Taken together, these findings provide some of the first evidence on how the choice between a centralized or decentralized structure of a tax authority impacts government fiscal capacity. I have shown that a more centralized structure entails certain costs for tax enforcement, particularly by

making it harder to collect local information. In the final part of the paper, I perform a back-of-the-envelope calculation comparing the foregone tax revenues to the estimated administrative savings generated by the reform. On average, I find that the reform resulted in more foregone revenues than savings. Future work should investigate whether the reform had countervailing, positive effects on tax enforcement in areas served by offices that absorbed tax agents from the closed locations. More generally, by shedding light on the mechanisms behind the impact of the reform, this paper suggests that policymakers interested in implementing a similar policy should consider how relevant acquiring local information is in their context and whether geographical frictions can make that process more difficult.

## Related Literature

**Related Literature** This paper relates and contributes to various strands of the literature. First, it speaks to the literature on tax administration in developing countries (Pomeranz and Vila-Belda 2019; Okunogbe and Tourek 2024). Past work has studied the role of incentives for tax collectors (Khan, Khwaja, and Olken 2016, 2019), the type of agent hired (Balán et al. 2022), specialization (Basri et al. 2021), and the potentialities of technology (Okunogbe and Santoro 2023; Dzansi et al. 2022; Knebelmann, Pouliquen, and Sarr 2023). This paper shows that the decision on how to organize the presence of tax authorities across a territory can affect their effectiveness.

Second, this paper is related to the literatures on the importance of third-party reporting (Pomeranz 2015; Naritomi 2019; Jensen 2022) and on the value of local information (Balán et al. 2022; Knebelmann, Pouliquen, and Sarr 2023) for taxation. It shows how a tax authority's ability to collect local information is constrained by the distance from taxpayers, and highlights that this ability is especially important in areas with limited third-party reporting.

Third, this paper contributes to the literature public sector personnel (Finan, Olken, and Pande 2017). While previous literature has investigated the role of selection (Dal Bó, Finan, and Rossi 2013; Ashraf et al. 2020), incentives (Bertrand et al. 2020), and managers (Fenizia 2022), this paper follows Besley et al. 2022 call to use microdata for examining the effects of system-wide reforms and the importance of different organizational designs. Since the reform examined here alters the size of tax jurisdictions, this paper also connects to the literature on decentralization which

explores the effects of different polity sizes (Bardhan 2002; Narasimhan and Weaver 2023; Dahis and Szerman 2024).

Finally, this paper speaks to the literature on the relationship between distance from capitals and state capacity (Mann 1984; Herbst 2014; Müller-Crepon 2023).By focusing on a specific dimension of state capacity and leveraging detailed information on the locations of bureaucrats performing this function, I am able to provide causal evidence on how state capacity deteriorates with distance.

The rest of the paper is organized as follows: Section **??** describes the context and the tax authority reform, while section **??** introduces the data. Section **??** presents evidence on the effects of the reform on tax revenues, whereas Section **??** investigate the mechanisms behinds these effects. In Section **??** I perform a back of the envelope calculation of the net effects of the before, before concluding in Section **??**.

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